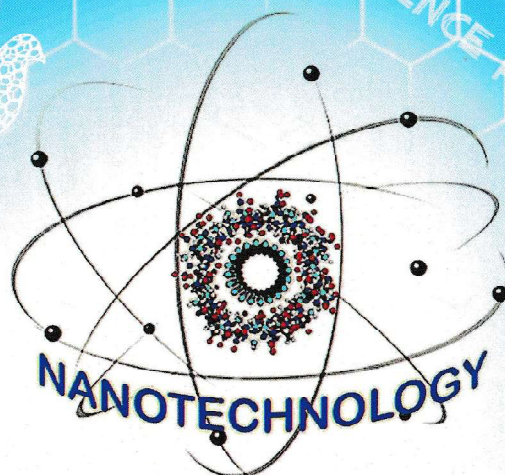




ABSTRACT BOOK

**International research
and practice conference:**

**NANOTECHNOLOGY
AND NANOMATERIALS
(NANO-2026)**

26-28 of August 2026
Chernivtsi, UKRAINE

**INTERNATIONAL RESEARCH
AND PRACTICE CONFERENCE
“NANOTECHNOLOGY
AND NANOMATERIALS”**

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Abstract book

The International research and practice conference “Nanotechnology and nanomaterials” (NANO-2026). Abstract Book of participants of the International research and practice conference, 26–28 August 2026, Chernivtsi. Edited by Dr. Olena Fesenko. – Kyiv: 2026. – P. 662.

This book contains the abstracts of contributions presented at the International research and practice conference “Nanotechnology and Nanomaterials” (NANO-2026), held on 26–28 August 2026 at Yuriy Fedkovych Chernivtsi National University, Chernivtsi, Ukraine.

The NANO-2026 Conference was organized by the Institute of Physics of NAS of Ukraine with the participation of the University of Tartu (Estonia), Yuriy Fedkovych Chernivtsi National University, University of Turin (Italy) and Pierre and Marie Curie University – Paris 6 (France).

NANO-2026 is the 14th conference in the series of NANO-conferences initiated by the Institute of Physics of NAS of Ukraine in 2012 in the framework of FP7 Nanotwinning project. From year to year, they attract more attention and participants. In 2012, the first meeting was held in the format of International Summer School for young scientists «Nanotechnology: from fundamental research to innovations». The 2013 and 2014 conferences were organized in conjunction with the International Summer Schools for young scientists under the same title. In 2013, this event was attended by more than 300 scientists, in 2014-2017, 450 scientists took part and in 2018 it gathered above 650 participants. In 2021 conference was attended by more than 700 scientists from Ukraine, Poland, Italy, Estonia, France, Austria, Germany, Greece, Turkey, USA, Romania, Moldova, Czech Republic, Taiwan, Lithuania, Egypt, India, Algeria, Indonesia and other countries. In 2021 - 2025 the Organizer Committee has received more than 800 application forms from about 25 countries of the world each years.

The NANO-2026 conference brought together leading scientists and young researchers from many countries of the world. This year its topics were as follows: Nanobiotechnology for health-care; Nanochemistry and biotechnology; Nanocomposites and nanomaterials; Nanooptics and photonics; Nanoplasmonics and surface enhanced spectroscopy; Nanoscale physics; Nanostructured surfaces; Physico-chemical nanomaterials science.

Website of the NANO-2026 conference: <http://nano-conference.iop.kiev.ua>

The effect of mechanical loading on the strength and electrical properties of polymer composites with nanocarbon fillers

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This work presents the results of studies of elastic and strength characteristics, as well as electrical resistance, under uniaxial mechanical loading of polymer composites (PCs) with different types of nanocarbon fillers. The composites were made using polymers such as epoxy resin ED20 and polychlorotrifluoroethylene (PCTFE, trademark fluoroplast-3). Thermally exfoliated graphite (TEG) and TEG modified with silicon oxide were used as fillers, as well as nanocarbon structures with lower *aspect ratios* obtained on the basis of TEG: doubly thermally exfoliated graphite and graphite nanoplates. Carbon fibers were also added to the composites to improve mechanical properties. The mechanical properties and electrical resistance of PCs under uniaxial compression were investigated using an automated installation based on the IMASH-20-78 serial equipment. The electrical resistance was measured using the two- or four-probe method.

The study demonstrates that the strength properties of PCs are significantly influenced by the temperature conditions during PCs production, the structural and morphological state of the carbon filler in PCi, its content, and how it is distributed within the polymer matrix. The main factors affecting the deformation of PCs and changes in electrical resistance were found to be the type and content of the carbon filler, the conductivity of the filler particles, and the electrical contact resistance between them, as well as the number of conductive filler particle chains formed in the composite. The strength characteristics of the polymer matrix also significantly impact the PCs overall mechanical properties.